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/* Creating a folder named "csp1018" and set up a library */
libname csp '/home/u64011409/csp1018';

proc contents data=csp.scr;
run;

/* Task 1.2: Selecting variables from 'scr' and create 'enroll' dataset */

data csp.enroll;
  set csp.scr; /* Load the 'scr' dataset */

  /* Keep only the necessary observations where Q26 (elig) is 3 */
  if Q26 = 3; /* Keep only enrolled individuals */

  /* Select and rename the required variables */
  keep center patient Q1 Q2date Q3 Q4 Q26 Q27date Q29date; /* Retain only necessary variables */

  /* Rename the variables */
  rename Q1 = gender
         Q2date = age
         Q3 = latino
         Q4 = race
         Q26 = elig
         Q27date = edate
         Q29date = ddate;
run;

/*1.3 Sorting the 'enroll' dataset by the patient variable */
proc sort data=csp.enroll;
  by patient; /* Sort by the patient variable */
run;

/* Sort the 'phys' dataset by the patient variable (although it indicates it is already sorted) */
proc sort data=csp.phys;
  by patient; /* Sort by the patient variable */
run;

/*Now proceed with the match-merge */

data csp.task1;
  merge csp.enroll(in=a) csp.phys(in=b); /* Merge datasets */
  by patient; /* Using the patient variable as the key */

  /* Keep only observations where individuals were enrolled */
  if a and b; /* Retaining only those in both datasets */
run;

/*Check the contents of the task1 dataset */

proc contents data=csp.task1;
run;

/1.4 Display the first 10 observations of the merged dataset to verify the data */

proc print data=csp.task1 (obs=10);
run;

/* Checking the contents of the task1 dataset */
proc contents data=csp.task1;
run;

/*////////////////////TASK 2////////////////////////////////////*/

/* Step 2.1: Create user-defined formats for gender and race */
proc format;
  value genderfmt
    1 = 'Male'
    2 = 'Female';

  value racefmt
    1 = 'White'

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2 = 'Black'
3 = 'Asian'
4 = 'Other';

run;

/* Step 2.2: Create a new dataset named 'task2' and select specified variables */

data csp.task2;
  set csp.task1; /* Read from the task1 dataset */

  /* Apply formats */
  format gender genderfmt. race racefmt.;

  /* Select variables*/
  keep CENTER PATIENT age gender race edate ddate
        Q1 Q2 Q3 Q4SYS Q4DIA
        Q5 Q6 Q7 Q8 Q9
        Q10 Q11 Q12 Q13 Q14 Q15 Q16 Q17; /* Keep only specified variables */

run;

/*Step 2.3:*/

data csp.task2;
  set csp.task2; /* Read from the task2 dataset */

  /* Convert height from inches to meters */
  height_m = Q1 * 0.0254;

  /* Convert weight from pounds to kilograms */
  weight_kg = Q2 * 0.453592;

  /* Calculate BMI */
  bmi = weight_kg / (height_m * height_m);

run;

/*Step 2.4:*/

data csp.task2;
  set csp.task2; /* Read from the task2 dataset */

  /* Create bmi_group variable based on bmi */
  if bmi < 18.5 then bmi_group = 1; /* Underweight */
  else if 18.5 <= bmi < 25 then bmi_group = 2; /* Normal */
  else if 25 <= bmi < 30 then bmi_group = 3; /* Overweight */
  else if bmi >= 30 then bmi_group = 4; /* Obese */

run;

/* Step 2.5: Create user-defined format for bmi_group */

proc format;
  value bmiformat
    1 = 'Underweight'
    2 = 'Normal'
    3 = 'Overweight'
    4 = 'Obese';

run;

/* Step 2.5: Apply the format to bmi_group */

data csp.task2;
  set csp.task2; /* Read from the task2 dataset */

  /* Apply the bmi_group format */
  format bmi_group bmiformat.;

run;

proc print data=csp.task2(obs=10);
  title 'First 10 Observations of Task2 Dataset';
run;

/*////////////////////TASK 3////////////////////////////////////*/

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/* Step 3.1: Transform edate and ddate from numeric to date values */

data csp.task3_temp; /* Create a temporary dataset to hold the transformation */
  set csp.task2; /* Read from the task2 dataset */

  /* Convert numeric dates to actual date values */
  format edate ddate date9.; /* Apply date9 format */

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/* Transform edate and ddate */
edate = intnx('day', '07MAY2000'd, edate); /* Transform edate */
ddate = intnx('day', '07MAY2000'd, ddate); /* Transform ddate */

run;

/*checking the transformation */

proc print data=csp.task3_temp (obs=10); /* Adjust obs=10 as needed */
run;

/*Step 3.2: Calculating lag and assign labels*/
data csp.task3_temp; /* Continue using the temporary dataset */
set csp.task3_temp; /* Read from the temporary dataset created in Task 3.1 */

/* Create lag variable */
lag = ddate - edate; /* Calculate the difference in days between ddate and edate */

/* Assign descriptive labels */
label edate = "Date of Entry"
      ddate = "Date of Diagnosis";
run;

/* Print the first few observations to check the new variables and labels */
proc print data=csp.task3_temp (obs=10); /* Adjust obs=10 as needed */
run;

/*Step 3.3: Create 'abnormalities' variable and new dataset 'task3'*/
data csp.task3; /* Create the final dataset */
set csp.task3_temp; /* Use the temporary dataset created in Task 3.2 */

/* Array to count abnormalities */
array q_vars{11} Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14 Q15 Q16 Q17; /* Adjust the array size according to the number of Q variables */
abnormalities = 0; /* Initialize the abnormalities counter */

/* Loop through the array to count the number of abnormalities (value of 2) */
do i = 1 to dim(q_vars);
    if q_vars{i} = 2 then abnormalities + 1; /* Increment counter for each value of 2 */
end;

/* Keep only the relevant variables */
keep edate ddate lag abnormalities;
run;

/* Step 3.4: Print out the first 20 observations of the task3 dataset */
proc print data=csp.task3 (obs=20); /* Print the first 20 observations */
    title "First 20 Observations of the Task 3 Dataset"; /*Optional: Add a title to the output*/
run;

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